Supplemental Notes for Further Data Analysis

Oct 30, 2026 · @Shahriar Akhavan Tafti

Purpose

This supplement collects the remaining analysis steps for the coherence project so they can be merged into the main analysis plan. It covers three things: follow-ups on the results that came out significant, a new within-region power analysis to address focal vs. distributed processing, and a request for a written methods description of the last run.

Follow-ups on the completed analysis

Transcribed from the handwritten meeting notes. Items marked (?) were hard to read and should be checked against the notebook.

1. **Confirm the GEE comparator (reference category).** By default the GEE uses either the first or the last level as the reference. Confirm which frequency bin is the comparator for the frequency factor (note reads "5 or 12 Hz" (?), i.e. the edges of the theta band).
2. **Directionality of significant results.** Run Granger causality only on the region pairs and bands that came out significant, to establish which region leads.
3. **Theta-gamma coupling.** Run cross-frequency (phase-amplitude) coupling, again restricted to the significant results.
4. **Side question (not for Joe).** Is there a gamma effect if every stimulus presentation is coded individually, rather than collapsing cue presentations?

Also noted at the top of the page, without context: **20 x 16 x 8 x 2** (?). Possibly the design dimensions or a cell count; confirm what this refers to.

New analysis: within-region theta and gamma power

Goal: test whether theta or gamma power goes up or down within each region across periods and sessions. This addresses the focal vs. distributed processing question and is separate from coherence (no between-region measure).

Data windows. 10 s baseline, 10 s Cue 1, 10 s Cue 2, on Day 1 and Day 4.

Dependent variable. Theta or gamma power/amplitude per region. Same frequency binning as the earlier band-profile analysis, but on power, not coherence.

Statistics. Each region is run independently. Region is not a factor (the shared sheet lists x 5 (region); that should be removed). Per region and per band:

Model	Design	DV
Band-level	2 (session: Day 1, Day 4) x 3 (period: Baseline, Cue 1, Cue 2)	Mean band power
Frequency-resolved	2 (session) x 3 (period) x F (frequency bin)	Power per frequency bin

Across 5 regions and 2 bands this is 20 models in total. Decide whether to correct for multiple comparisons across regions.

Figures. 6 per region: 3 theta, 3 gamma. Assumed layout per band (confirm): one band-level plot (session x period), plus one frequency profile for Day 1 and one for Day 4, each showing the three periods.

Y-axis units. The example sheet's units look wrong. The correct label depends on how power is computed:

Method	Unit
Spectral power density (PSD, e.g. Welch)	$\mu V^2/Hz$
Log-scaled power	dB
Hilbert envelope amplitude	μV
Normalized to baseline	% change or z-score (unitless)

If the data are normalized per rat (next section), the axis becomes % change or z, not μV.

Normalizing each rat

Raw power varies a lot between animals (electrode impedance, placement), so one high-power rat can drive or mask a group effect. Normalize within each rat before collapsing across animals. Options:

- **Baseline normalization (preferred fit for this design):** express Cue 1 and Cue 2 as % change or dB relative to that rat's baseline, separately for Day 1 and Day 4. Downside: baseline drops out as a level of the period factor.
- **Within-rat z-score:** z-score each rat's power across all its windows (both days, all periods). Keeps all 6 cells in the 2 x 3 design.
- **Relative power:** band power divided by total power (e.g. 1-100 Hz) in the same window. Controls for broadband shifts.

Whichever is chosen, keep rat as the subject/cluster variable in the model, and check results with and without any outlier rat.

Request: written methods for the last run

For the final run, have the pipeline output a detailed written description alongside the results, so the methods can be reported accurately. It should state:

- Data included: rats (n), sessions, regions, channels per region, window lengths and alignment
- Preprocessing: filtering, referencing, artifact rejection, downsampling
- Spectral method: function used, window/taper, overlap, frequency resolution, band edges for theta and gamma
- The exact DV (coherence or power) and its units
- Normalization applied, if any
- Statistical model: test type, factors and levels, reference category for each factor, working correlation, subject/cluster variable, post hoc comparisons and correction method
- Which comparisons were significant, with test statistic, df and p value

Open questions and action items

- [ ] Confirm the GEE reference category for frequency (5 or 12 Hz?)
- [ ] Run Granger directionality on significant pairs
- [ ] Run theta-gamma coupling on significant results
- [ ] Test the per-stimulus gamma coding (side question)
- [ ] Confirm what "20 x 16 x 8 x 2" refers to
- [ ] Run within-region power: 5 regions x 2 bands, 2 x 3 and 2 x 3 x F per region

- [ ] Pick the power measure, then fix y-axis units on the example graphs
- [ ] Choose the per-rat normalization method
- [ ] Confirm the 6-figure layout per region
- [ ] Add written methods output to the final run
- [ ] Attach the example-graph sheet to this doc (it wasn't included here)
- [ ] Optional: Zoom or call to talk through the design